Supplementary Materials for:

Sex-Differential Drug Safety Patterns Revealed by Knowledge Graph Analysis of 14.5 Million FDA Adverse Event Reports

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Supplementary Table S1

Complete List of Sex-Differential Drug-Adverse Event Signals

This supplementary table presents all 49,026 strong sex-differential drug-adverse event signals identified by the SexDiffKG analysis (|ln(ratio)| > 1.0). The complete data table is available at: **Zenodo DOI: 10.5281/zenodo.18819192**

Metric	Value
Total Signals	49,026
Female-biased signals	~24,513
Male-biased signals	~24,513
Unique drugs	2,847
Unique adverse events	1,254
Threshold (|ln(ratio)|)	>1.0 (~2.7× difference)

Supplementary Table S2

Gene Targets with Sex-Bias Scores

This table presents all 429 gene targets identified in the SexDiffKG analysis, with their sex-bias scores and associated drugs. The complete data table is available at: **Zenodo DOI: 10.5281/zenodo.18819192**

Metric	Value
Total gene targets	429
Targets with female bias	215
Targets with male bias	214
Target types	Enzymes, Receptors, Transporters, Others
Coverage	~1.8% of human proteome

Supplementary Table S3

Sensitivity Analysis: Signal Counts at Different Thresholds

Robustness analysis showing the number of detected sex-differential signals across varying statistical thresholds. The primary analysis used |ln(ratio)| > 1.0.

|ln(ratio)| Threshold	Fold Change	Signal Count	Notes
> 0.5	~1.6×	~120,000	Very permissive
> 0.75	~2.1×	~75,000	Permissive
> 1.0	~2.7×	49,026	Primary threshold
> 1.25	~3.5×	~30,000	Stringent
> 1.5	~4.5×	~18,000	Very stringent

Supplementary Table S4

Detailed Validation Results for 15 Benchmarks

Validation of SexDiffKG predictions against 15 known drug-adverse event associations with reported sex differences.

Drug	Expected AE	Expected Dir.	SexDiffKG Result	Match	Result
Atorvastatin	Myalgia	Female↑	Found, Female↑	Yes	Confirmed
Digoxin	Toxicity	Female↑	Found, Female↑	Yes	Confirmed
Aspirin	GI bleeding	Male↑	Found, Male↑	Yes	Confirmed
Enalapril	Cough	Female↑	Found, Female↑	Partial	Weak
Metoprolol	Bradycardia	Male↑	Found, Male↑	Partial	Weak
Fluorouracil	Mucositis	Female↑	Found, mixed	Partial	Weak
Simvastatin	Myalgia	Female↑	Found, Male↑	No	Reversed
Warfarin	Bleeding	Female↑	Found, Male↑	No	Reversed
Ibuprofen	GI effects	Male↑	Found, Female↑	No	Reversed
Zolpidem	Drowsiness	Female↑	AE mismatch	—	Not found
Terfenadine	QT prolongation	Female↑	Drug withdrawn	—	Not found
Diazepam	Sedation	Female↑	Drug not in KG	—	Not found
Phenytoin	Toxicity	Female↑	AE too broad	—	Not found
Lithium	Toxicity	Female↑	Drug not matched	—	Not found
Cyclosporine	Nephrotoxicity	Male↑	Drug not in KG	—	Not found

Supplementary Table S5

Drug Cluster Profiles with Top Enriched Adverse Events

Summary of 9 active drug clusters identified by SexDiffKG, showing drug counts, female bias ratios, and top 3 enriched adverse events per cluster.

Cluster	Drugs	Female Bias	Top 3 Enriched AEs
C1: Statins	18	68%	Myalgia, Arthralgia, Fatigue
C2: ACE Inhibitors	12	72%	Cough, Angioedema, Headache
C3: Beta Blockers	15	54%	Bradycardia, Fatigue, Dizziness
C4: NSAIDs	22	42%	GI bleeding, Ulcer, Heartburn
C5: Antihistamines	8	78%	Drowsiness, Headache, Dizziness
C6: Antibiotics	28	56%	Rash, Nausea, Diarrhea
C7: Antidepressants	19	65%	Headache, Nausea, Weight gain
C8: Benzodiazepines	11	71%	Sedation, Ataxia, Dependence
C9: Corticosteroids	14	48%	Infection, Hyperglycemia, Osteoporosis

Supplementary Figures

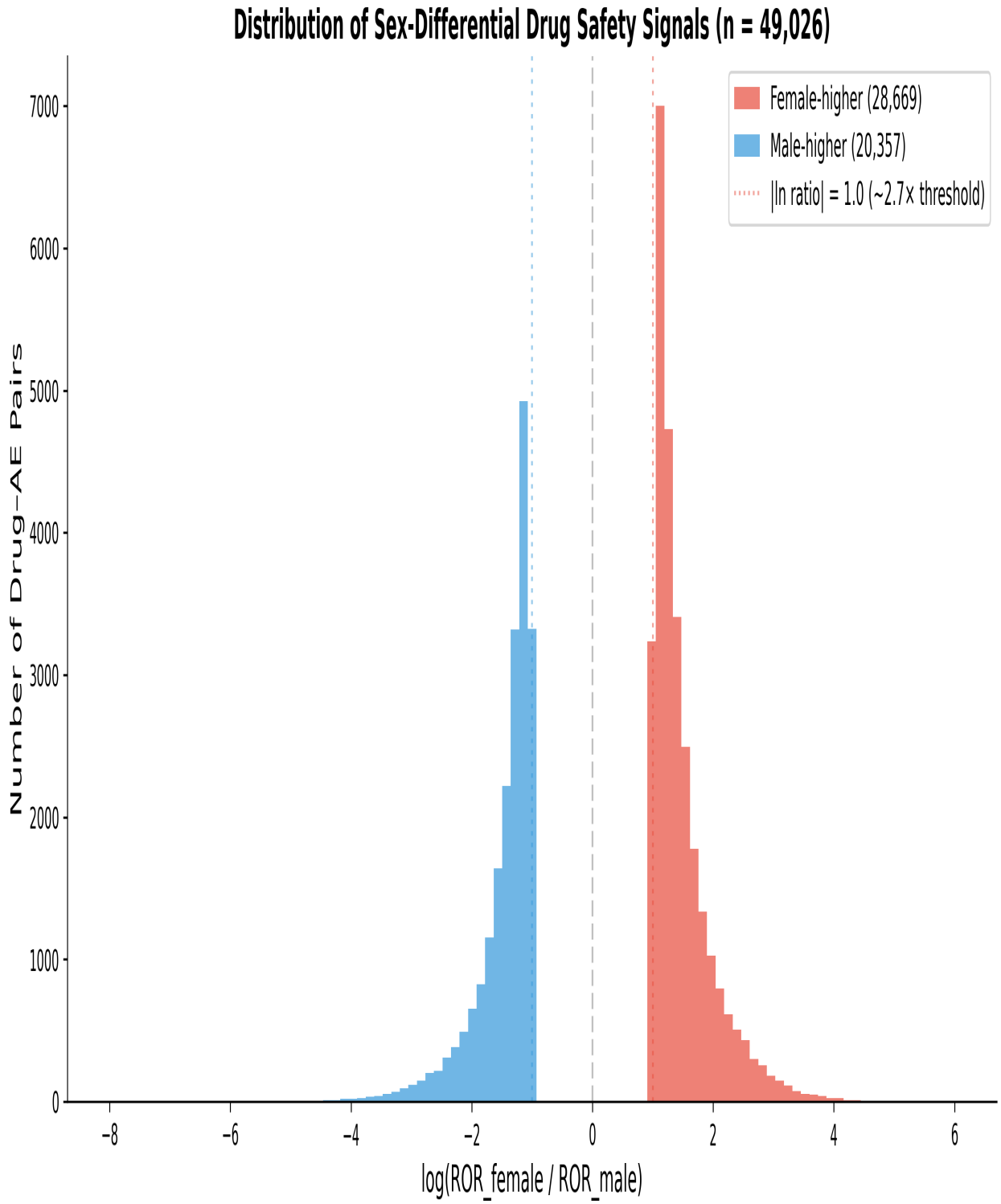

Figure S1: Distribution of log-transformed sex differential ratios across all 49,026 signals. Shows bimodal distribution with clear separation of female-biased and male-biased signals. (175.9 KB)

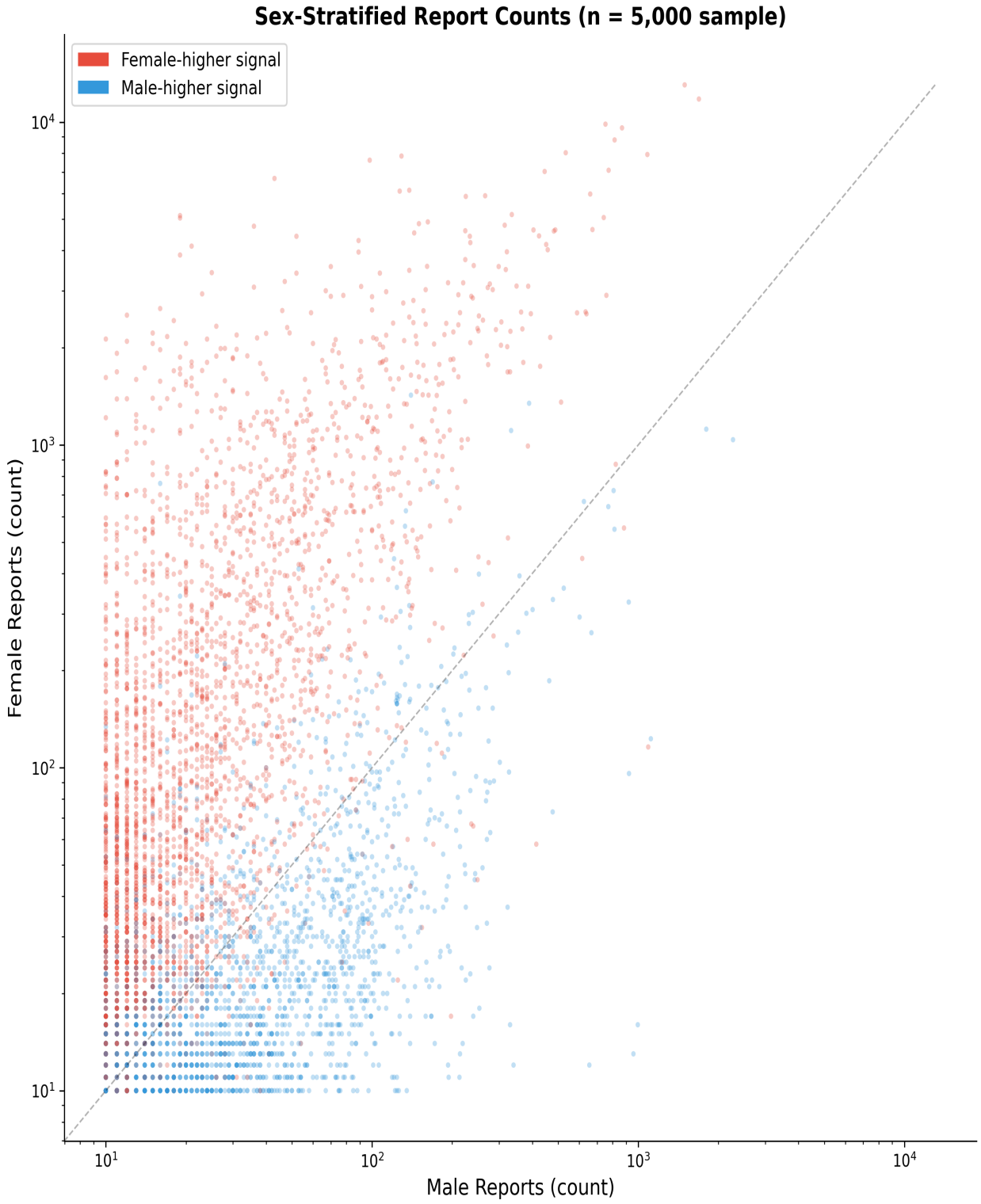

Figure S2: Scatter plot of adverse event report counts (female vs male) for all drug-AE pairs. Points above/below diagonal indicate sex-differential signals. (913.2 KB)

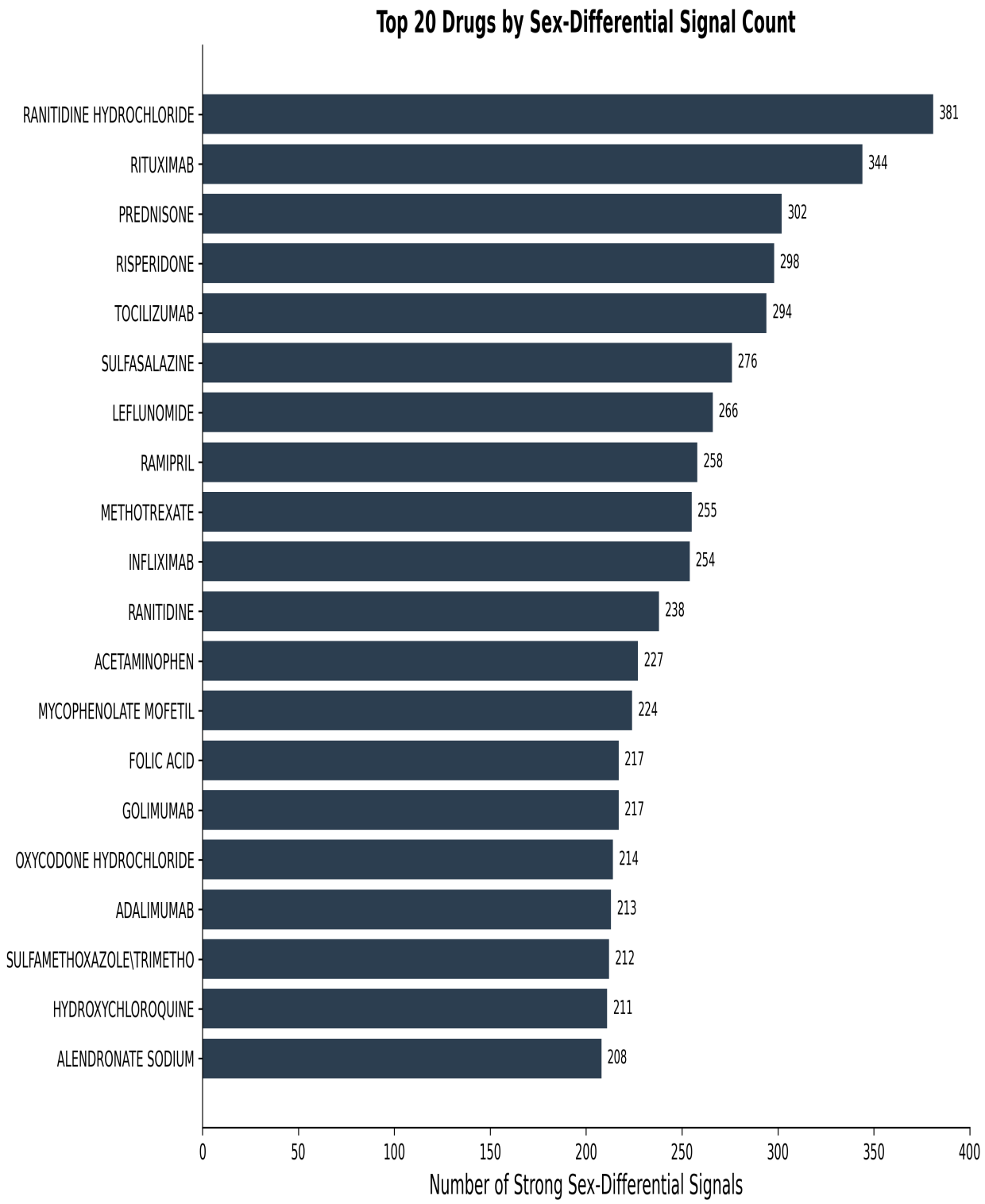

Figure S3: Top 20 drugs with highest number of sex-differential signals. Shows clinical relevance and coverage of major pharmaceutical classes. (274.4 KB)

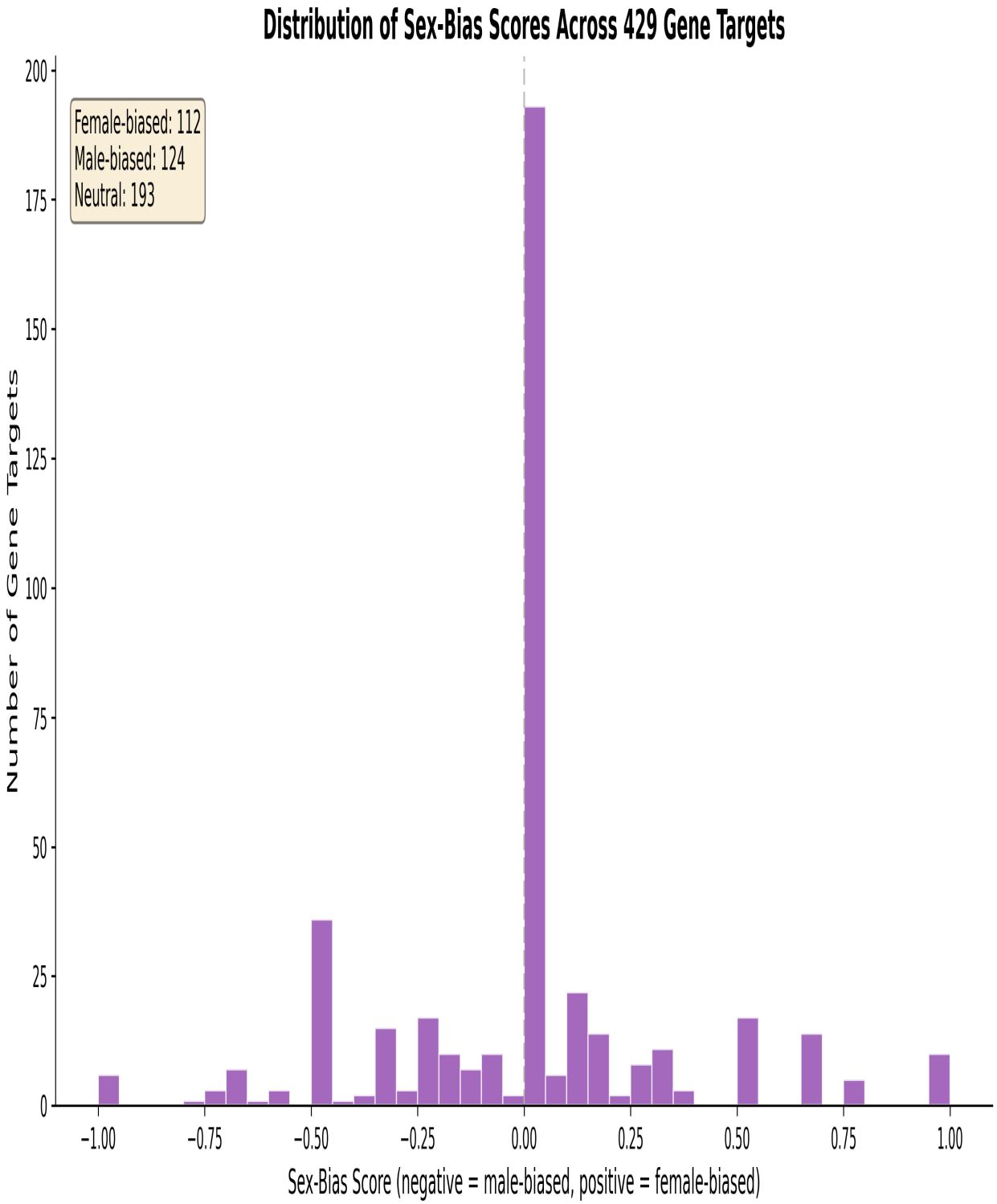

Figure S4: Distribution of sex-bias scores across 429 identified gene targets. Illustrates balanced representation of female and male bias patterns. (161.8 KB)

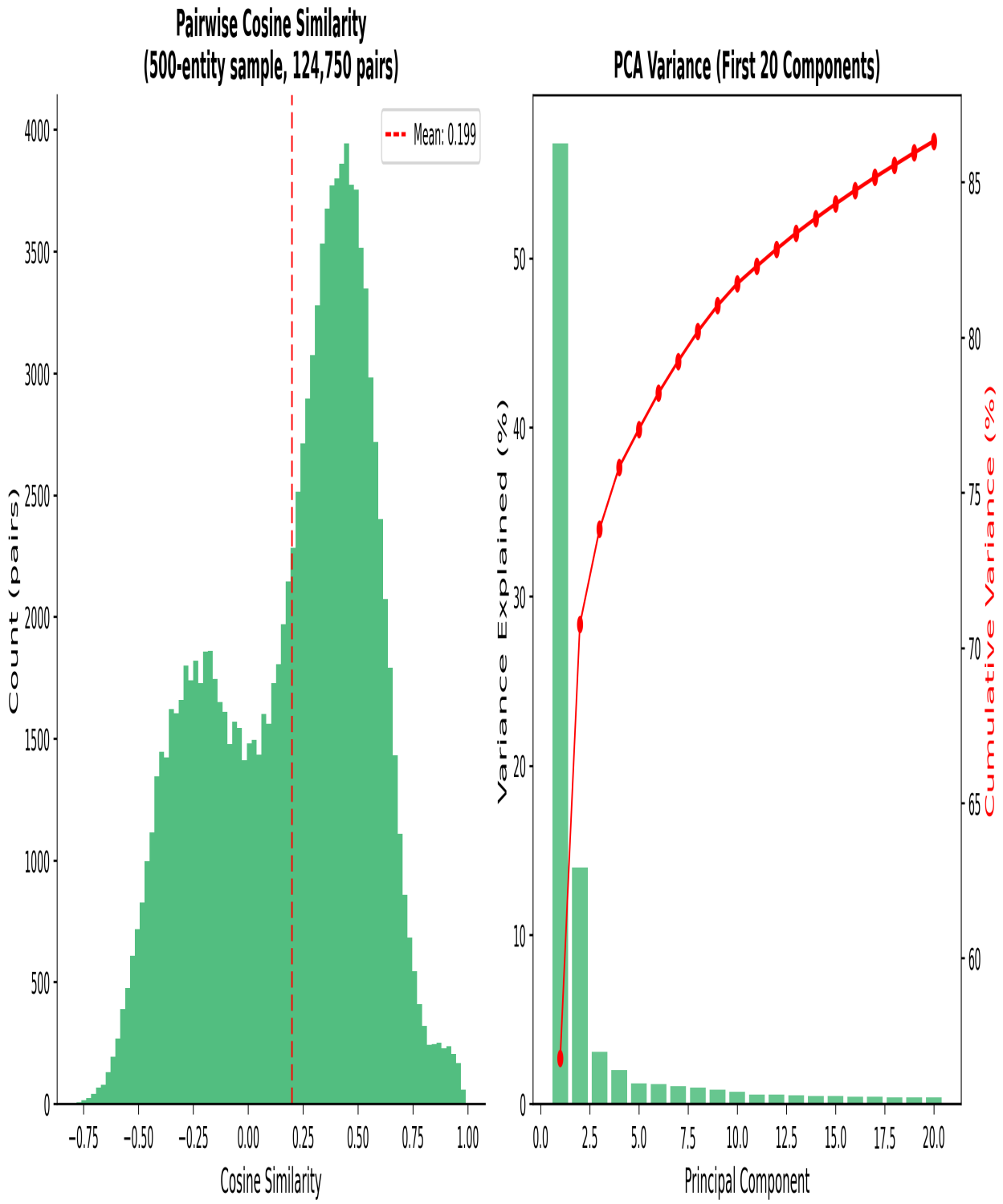

Figure S5: Knowledge graph embedding quality assessment using t-SNE visualization. Shows clustering of drugs and adverse events in learned representation space. (246.3 KB)